SPOTIFY ANALYSIS

1. Project Overview:

This project focuses on performing an end-to-end data analysis using Spotify history data. Includes Spotify track URI, timestamp (TS), platform, track name, artist name, album name, and usage metrics like shuffle and skip status. **The goal**:

- Analyzes albums, artists, and tracks to visualize listening trends over time, identify top performers, and calculate year-on-year growth
- Implements quadrant analysis to categorize tracks based on listening frequency and duration.
- Utilizes advanced features like drill-through, drill-down, and hierarchical navigation to allow users to export specific, granular data for stakeholders.

2. Dataset Summary:

- Rows: 149,861

- Columns: 11

- Key Features:

+ **Essential Metadata**: Each entry includes a unique track identifier (Spotify track URI), specific timestamps for listening sessions, as well as data on platform, track names, artist names, and album names.

+ **Usage Insights**: It captures behavioral metrics such as shuffle mode status and skip events, which are critical for calculating user engagement, popularity, and drop-off rates

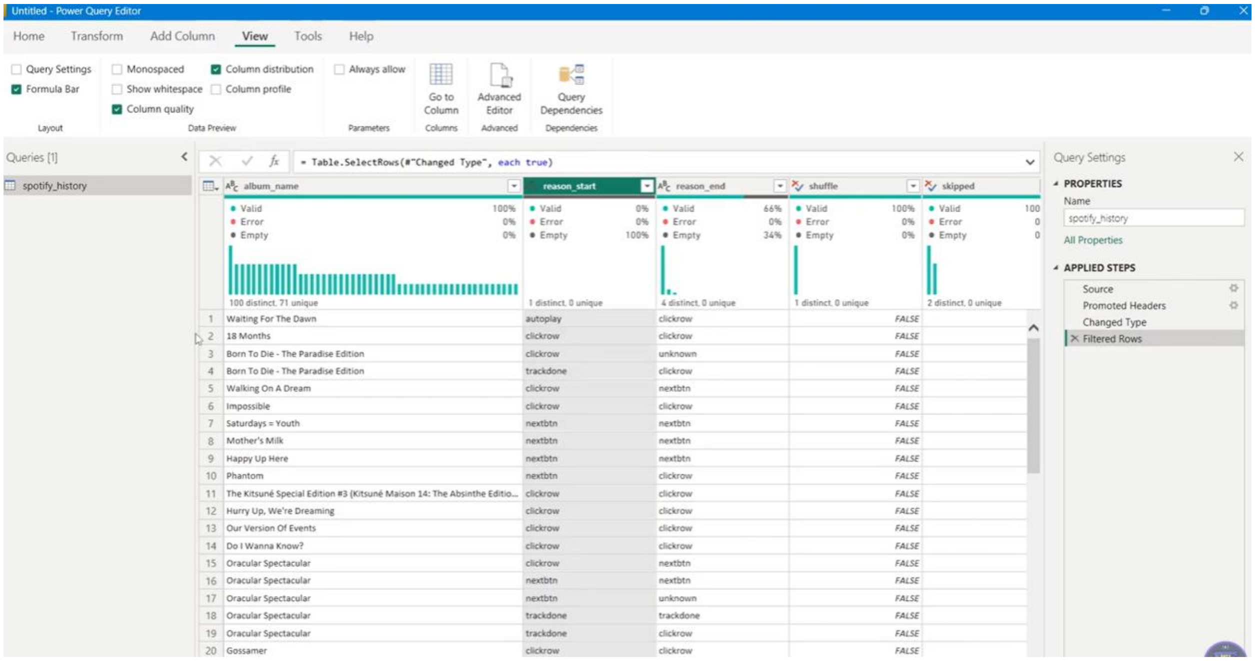
+ **Data Integrity**: The dataset is described as being high-quality and pre-cleaned, requiring minimal transformation to perform complex analytical tasks like quadrant and time-intelligence analysis

- Error Data: 2 values and no missing data

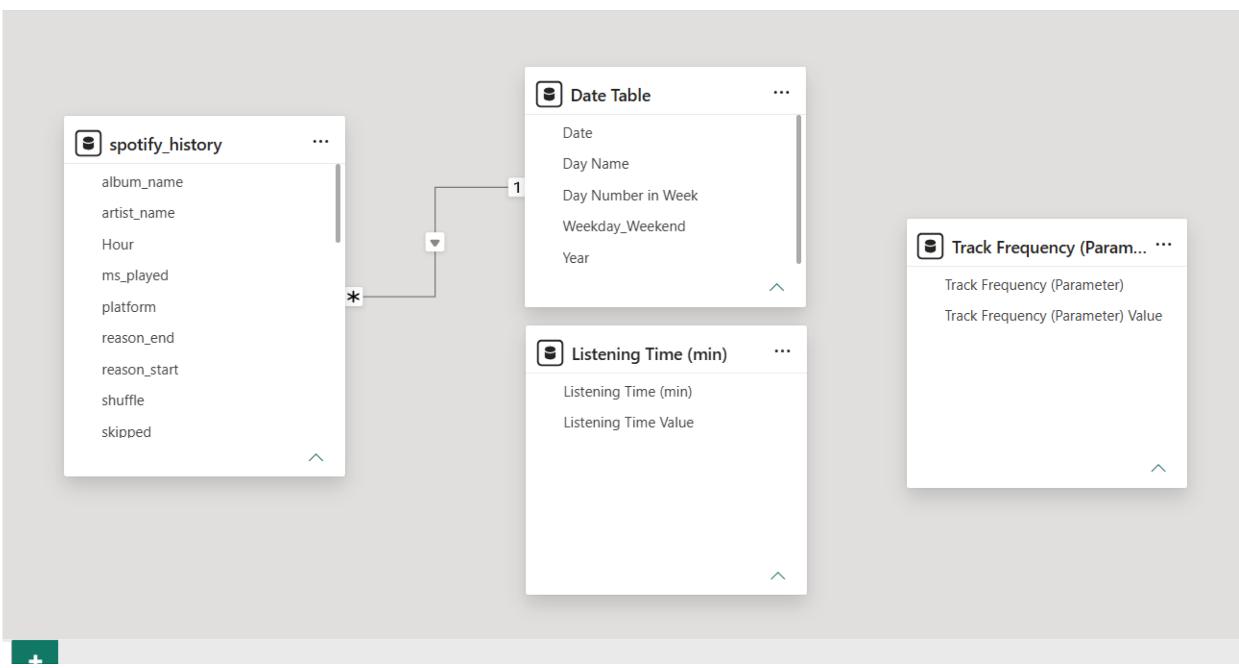
3. Exploratory Data Analysis using Power BI:

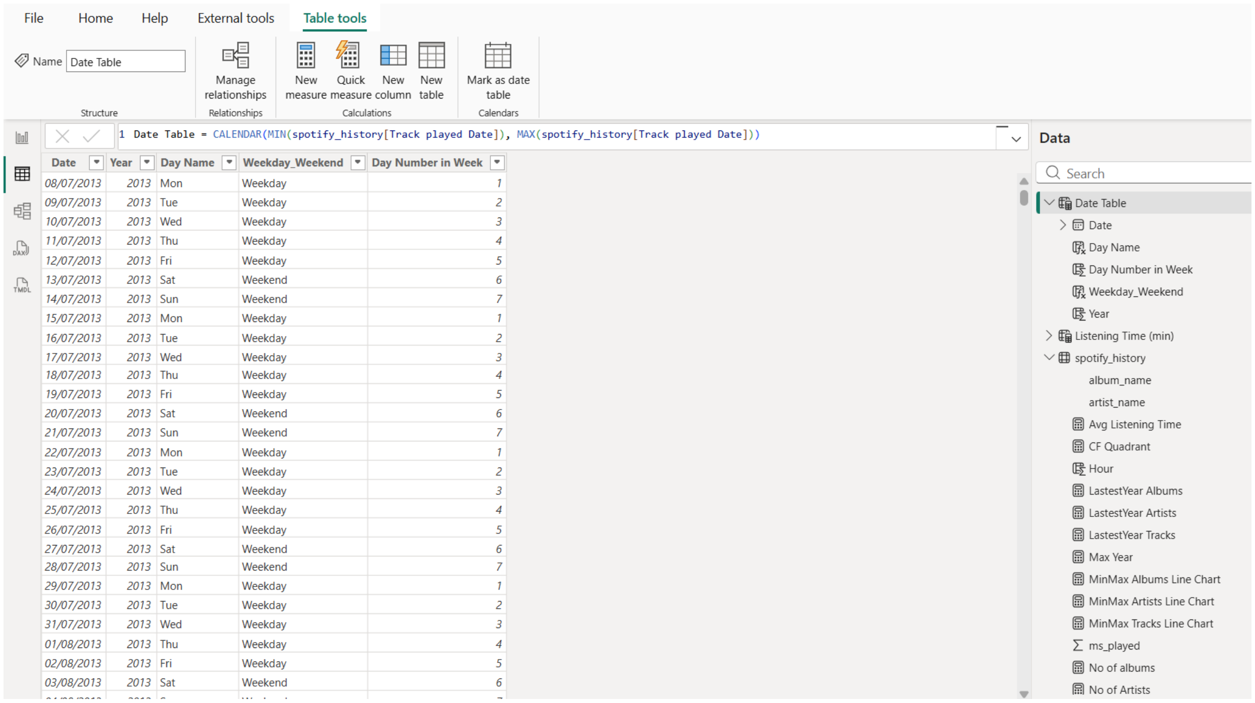
We began with data preparation and cleaning in Power BI:

- **Data Preparation & Transformation**: Using Power Query Editor to perform ETL operations, checking for data quality, and ensuring there are no errors.



- **Data Modeling**: Establishing relationships between the main **Spotify history** data and a custom **Calendar Table** (date table). This is crucial for enabling **Time Intelligence** functions.

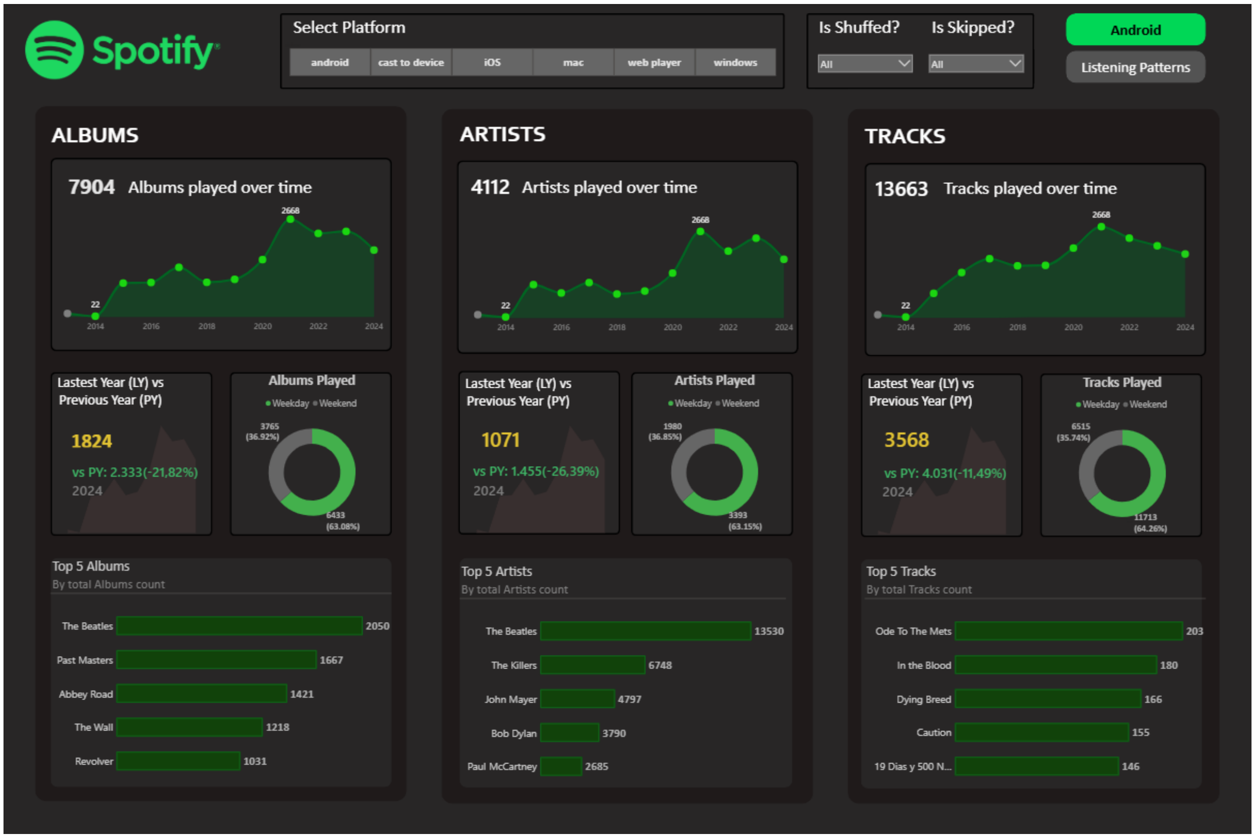




- **DAX Calculations:** Creating advanced measures to derive insights, details in Query_spotify.dax

4. Dashboard in Power BI:

Finally, we built an interactive dashboard in Power BI to present insights visually.


Spotify
Select Platform
android
cast to device
iOS
mac
web player
windows
Is Shuffed?
Is Skipped?
All
All
Android
Listening Patterns
ALBUMS
7904 Albums played over time
2668
22
2014
2016
2018
2020
2022
2024
Lastest Year (LY) vs Previous Year (PY)
1824
vs PY: 2.333(-21,82%)
2024
Albums Played
Weekday Weekend
3765 (36.92%)
6433 (63.08%)
Top 5 Albums
By total Albums count
The Beatles 2050
Past Masters 1667
Abbey Road 1421
The Wall 1218
Revolver 1031
ARTISTS
4112 Artists played over time
2668
22
Lastest Year (LY) vs Previous Year (PY)
1071
vs PY: 1.455(-26,39%)
2024
Artists Played
Weekday Weekend
1980 (36.85%)
3393 (63.15%)
Top 5 Artists
By total Artists count
The Beatles 13530
The Killers 6748
John Mayer 4797
Bob Dylan 3790
Paul McCartney 2685
TRACKS
13663 Tracks played over time
2668
22
Lastest Year (LY) vs Previous Year (PY)
3568
vs PY: 4.031(-11,49%)
2024
Tracks Played
Weekday Weekend
6515 (35.74%)
11713 (64.26%)
Top 5 Tracks
By total Tracks count
Ode To The Mets 203
In the Blood 180
Dying Breed 166
Caution 155
19 Dias y 500 N... 146


Spotify
Year
Is Shuffed?
Is Skipped?
All
All
All
Listening Hour vs Day
Hour
Mon
Tue
Wed
Thu
Fri
Sat
Sun
Total
0 4015
1 3985
2 3617
3 3517
4 3195
5 3341
6 3452
7 2783
8 1665
9 1299
10 951
11 746
12 579
13 1205
14 1754
15 2147
16 3067
17 3634
18 3681
19 3503
20 3907
21 3723
22 3533
23 3837
Avg Listening Time (min) vs Track Frequency
Listening Time (min)
3
Track Frequency (Parameter)
90
Track Frequency
200
150
100
50
0
0
5
10
15
20

track_name	No of albums	No of Artists	No of Tracks	MilliSeconds Played	Avg Listening Time
✨	1	1	1	159099	0.88
!!!!!!!	1	1	1	13561	0.23
"45"	1	1	1	2726197	1.82
"C" Jam Blues - Live At Symphony Hall, Boston, MA/1947	1	1	1	231690	3.86
"Heroes" / "Helden" - 2001 Remaster	1	1	1	4946	0.08
"Hit the Quan" #HTQ	1	1	1	215516	1.20
"I Know"	1	1	1	138473	2.31
"Mucho, mucho" (Muñequita Linda)	2	1	1	232439	1.94
"Siboney"	1	1	1	36960	0.62
"Volver" - Arr. Roberto Pansera	1	1	1	12618	0.21
"Was He Slow?" - Music From The Motion Picture Baby Driver	1	1	1	106880	1.78
#1	2	1	1	2870560	3.42
#41 - Live at Wrigley Field, Chicago, IL - September 2010	1	1	1	967386	16.12
#Idiota	1	1	1	183903	3.07
#SELFIE	1	1	1	4563	0.08
(Are You) the One That I've Been Waiting For? - 2011 Remastered Version	1	1	1	245466	4.09
(Drill Sergeant)	1	1	1	45287	0.25
(Everybody Wanna Get Rich) Rite Away	1	1	1	491799	2.73
(Ghost) Riders in the Sky	1	1	1	216686	1.81
(I Can't Get No) Satisfaction - Live At University Of Leeds / 1971	1	1	1	149211	2.49
(I Can't Get No) Satisfaction - Mono Version	1	1	1	15135528	2.97
(I Just) Died In Your Arms	1	1	1	280400	4.67
(I Would Do) Anything For You	1	1	1	76765	1.28
(I'm Gonna) Love Me Again	1	1	1	250196	4.17
(I've Had) The Time of My Life - From "Dirty Dancing" Soundtrack	1	1	1	12704	0.21
(JFK)	1	1	1	109240	0.91
(levitation)	1	1	1	8156	0.14
(Nice Dream)	1	1	1	2752056	1.53
Total	7904	4112	13663	19039258100	2.13

5. Business Recommendations:

- **Marketing Adjustments**: If an album shows a decreasing trend in listening frequency, the business could increase marketing costs to boost user engagement.
- **Pricing Strategies**: The business could consider reducing premium charges or offering free trial periods (e.g., one or two months free) to encourage more users to listen, potentially increasing charges later.
- **Data-Driven Management**: By using the 'Details' tab to export underlying data, team members can present granular information to higher management to support informed business decision-making.